

MID TERM EXAMINATION
B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
 2nd Semester, May 2023

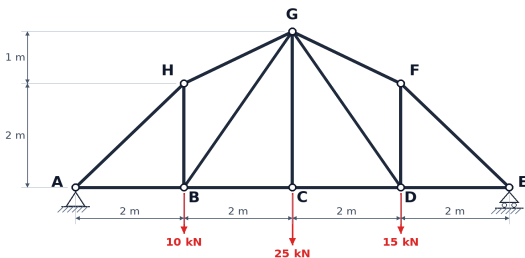
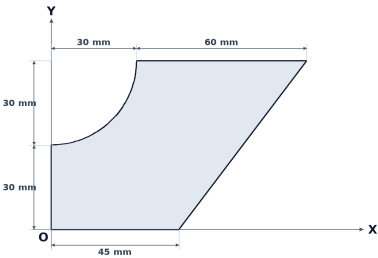
Paper Code: ES 114

Subject: Engineering Mechanics

Time: 1.5 Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	Write short notes on the following: a) State Lami's theorem. b) Perfect Truss assumptions c) State Varignon's theorem. d) What is a self-locking machine?	2.5 2.5 2.5 2.5	CO1 CO2 CO1 CO2
2	Derive the expression for the ratio of tensions for a flat belt passing over a pulley when it is just on the point of slipping.	10	CO2
3	Determine the force in GH, GD and GF member of truss as shown in figure. 	10	CO2
4	Find the centroid of given shaded area as shown in figure. 	10	CO1